

Against Causation

*A Formal Argument That Causality Is a Compression Artifact
of Bounded Observers, Not a Feature of Reality*

A Perspectival-Relational Account — Revised

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Abstract

We argue that causation is not an intrinsic feature of mind-independent reality but a relational structure that emerges at the interface between a system’s conditional independence structure and a bounded observer’s compression requirements. Drawing on Woodward’s interventionist semantics, the algorithmic independence of causal conditionals (Janzing and Schölkopf 2010), Price’s perspectivalism, and the time-symmetry of fundamental physics, we develop a three-level ontology: (i) the territory supplies undirected conditional independence structure via locality and symmetry; (ii) the observer contributes directionality, scale, and system boundaries; (iii) causation is the product of (i) and (ii) jointly—real at the relational level, absent at either level alone. We distinguish this position from both eliminativism and standard realism, clarify its ontological commitments, and develop its consequences for causal pluralism, the metaphysical privilege of fundamental physics, and the problem of mental causation.

1. Introduction

Causation is the organizing principle of empirical science. Every experiment seeks causes; every explanation invokes them; every intervention presupposes them. Yet the concept sits uneasily at the foundations. The fundamental equations of physics—the Schrödinger equation, the Einstein field equations, the Standard Model Lagrangian—are time-symmetric. The word ‘cause’ does not appear in them. As Russell (1913) observed, causation is conspicuously absent from the vocabulary of advanced physics. More recently, Norton (2003) has reinforced this point by cataloguing the persistent failures to extract a robust notion of cause from fundamental law.

This paper develops a perspectival-relational account of causation. The thesis: causation is the minimal directed graph that a bounded observer constructs to compress the correlational structure of its environment for the purpose of prediction and intervention. It is real—but real in the way that temperature is real: as a

macroscopic, frame-dependent quantity that exists at the interface between observer and system, not as a mind-independent feature of ultimate reality.

The account differs from eliminativism (Russell 1913; Norton 2003) in granting causation relational reality, and from standard interventionism (Woodward 2003) in denying that causal structure is a unique, observer-independent feature of the world. It is closest to the agent-based perspectivalism of Price (2007) and Price and Ramsey (2007), but grounds the observer's contribution in information-theoretic compression rather than in agency per se, and draws explicitly on the structural realism of Ladyman and Ross (2007) to characterize what the territory provides independently of any observer.

2. The Core Argument

2.1 The Formal Skeleton

The argument proceeds from four premises:

P1 (Boundedness). A bounded observer cannot store or process the full joint probability distribution over all variables in its environment. It requires compression.

P2 (Optimality of DAGs). A directed acyclic graph (DAG) satisfying the causal Markov condition is the most efficient compression scheme for statistical dependencies—the insight formalized by Bayesian networks (Pearl 2009).

P3 (Exhaustion). Once one has specified (a) the territory's undirected conditional independence structure and (b) the observer's contribution of directionality, scale, and system boundaries, there is no further fact about causation left to discover. (Argued in §2.4 below.)

P4 (Consequence). Therefore, causation is a relational structure at the observer–system interface. Without the territory's structure, the observer has nothing to compress. Without the bounded observer, the structure has no direction and no causal content.

An unbounded observer—Laplace's demon—with access to the complete joint distribution over all variables would have no use for causal arrows. It could read off any conditional probability directly. Causal structure, on this view, is what the world's correlational skeleton looks like to a finite agent that must compress in order to act.

2.2 Evidence from Physics

The time-symmetry of fundamental law provides evidence for the thesis. The asymmetry of 'cause precedes effect' is not written into the dynamical equations; it enters through thermodynamic boundary conditions—specifically, the low-entropy initial state of the universe (the 'Past Hypothesis' of Albert 2000)—and through the observer's coarse-graining choices. Both are arguably observer-relative: the selection of macrostates depends on the observer's resolution, and the thermodynamic arrow is a statistical artifact of that selection (Loewer 2012).

2.3 Information-Theoretic Corroboration

When one formalizes ‘compression of correlations,’ one arrives at quantities like Schreiber’s (2000) transfer entropy and Granger (1969) causality. Both are defined relative to an observer’s variable partition: change the partition, change the ‘causes.’ This is what the thesis predicts: causal attribution is partition-relative.

An important caveat: transfer entropy and Granger causality measure *predictive information transfer*, not intervention-supporting structure. They can detect spurious ‘causes’ driven by confounders (Lizier and Prokopenko 2010). We do not claim these measures constitute a full analysis of causation. Rather, they illustrate a specific feature of the thesis: that even our best formal proxies for causal attribution are explicitly observer-relative. The richer information-theoretic analysis comes from the algorithmic independence of causal conditionals (Section 2.4).

2.4 The Argument for P3: Nothing Left Over

P3—the claim that once we have the territory’s undirected structure plus the observer’s contribution, there is no residual causal fact—is the thesis’s central and most contested premise. It requires direct argument, which we now provide.

The Woodwardian challenge. Woodward (2003) argues that causation is grounded in invariant relationships under intervention: X causes Y if and only if there exists an intervention on X that changes Y . This seems to locate causation in the territory, not in the observer. However, the Woodwardian framework implicitly presupposes a bounded observer at three points: (i) the choice of which variables to define (the ‘variable partition’), (ii) the designation of what counts as an ‘intervention’ (requiring a system/environment boundary), and (iii) the temporal direction in which change is tracked. None of these is given by physics alone. Woodward’s invariance relations are real—but they are real relative to an agent’s partition, boundaries, and temporal orientation. The detailed conditions Woodward (2003, ch. 2–3) articulates for what constitutes an intervention—modularity, independence of the intervention variable from confounders, exogeneity—each require a prior decomposition of the world into distinct causal modules, which is itself a perspectival act.

Algorithmic independence as evidence. Janzing and Schölkopf (2010) show that for the true causal direction $X \rightarrow Y$, the mechanism $P(Y|X)$ is algorithmically independent of the cause distribution $P(X)$ —their descriptions share no mutual algorithmic information. This is a powerful result, and one might think it refutes our thesis by grounding causal direction in observer-independent algorithmic structure.

But notice what the result actually establishes: it provides a criterion for selecting among directed graph representations, given that one is constructing a directed graph in the first place. The undirected conditional dependence between X and Y is the territory’s contribution; the algorithmic independence criterion tells the observer *which direction to orient the edge*. This is precisely the observer–territory decomposition we propose. The criterion is a compression-quality principle—it identifies which orientation yields the most modular, compressible factorization—not evidence of mind-independent directed structure. Indeed, algorithmic independence is defined relative to a description language and a notion of Kolmogorov complexity, both of which involve a reference machine.

The exhaustion argument. Consider what a complete description of a physical system would include: the full quantum state, the Hamiltonian, and the boundary conditions. From these, all correlations and conditional independencies can in principle be derived. But no directed causal graph can be derived without

additionally specifying (a) a coarse-graining (variable partition), (b) a system/environment boundary (what is exogenous), and (c) a temporal asymmetry (arrow of time). These three specifications are contributions of the observer's perspective, not features of the quantum state. Once they are supplied, the causal DAG is fully determined by the resulting conditional independence structure plus the algorithmic independence criterion. Nothing is left over. This is what P3 asserts.

3. From Eliminativism to Perspectivalism

The raw argument, taken at face value, could yield eliminativism: causation is an illusion. We reject this reading in favor of a perspectival-relational alternative, for three reasons.

3.1 The Interventionist Objection

If one cuts a wire, the light goes off. This is not merely a compression convenience—it is a counterfactual fact about the system. The asymmetry between *do*(*X*) and *observe*(*X*) (Pearl 2009) has genuine predictive content.

The eliminativist must say that the intervention is itself a bounded observer constructing a subsystem boundary and labeling one side 'exogenous.' This is defensible but strained. Woodward (2003, ch. 3) develops in detail the conditions under which an act counts as an 'intervention,' and these conditions—modularity, independence, exogeneity—each involve variables and system boundaries that, as we argued in §2.4, presuppose an observer's perspective.

The perspectival account handles this naturally: the wire-cutting relationship is real at the observer–system interface. The intervention asymmetry is not merely epistemic; it reflects a genuine feature of how a bounded agent is embedded in the physical world. But it is relational rather than absolute. A differently bounded agent—one that can only intervene on electromagnetic fields, not on wires—would represent the same physical situation with a different causal graph, and both would be correct relative to their intervention capacities.

3.2 The Convergence Problem

Different observers with different compression needs and sensory apparatus often converge on roughly the same causal graphs. A neuroscientist and a physicist examining the same neuron arrive at compatible causal structures. If causation were purely an artifact of each observer's idiosyncratic compression, this convergence would be inexplicable.

This is a real challenge, and we take it seriously. Our response requires a precise decomposition of what is observer-contributed and what is territory-contributed.

The territory contributes *undirected conditional independence structure*: which variables are dependent on which, once others are conditioned upon. This structure is observer-independent in the sense that it is fixed by the physics. It is this structure that explains convergence: observers converge because they are all compressing the *same* underlying skeleton of dependencies.

What the observer contributes is threefold: (a) *directionality*—the arrow from cause to effect, which depends on the observer's temporal embedding and thermodynamic asymmetry; (b) *scale*—the choice of

coarse-graining level, which determines which variables exist in the causal model; and (c) *system boundaries*—what counts as exogenous versus endogenous, which shapes the intervention structure.

One might object: if the territory’s structure does the explanatory work in accounting for convergence, perhaps the structure *is* the causation, and the observer is merely accessing it. We resist this inference. The undirected structure alone does not constitute causation in any recognizable sense: it lacks direction, it lacks the do/observe asymmetry, and it does not support counterfactual reasoning. These features—the features that make causation causation—are observer-contributed. Convergence is explained by the shared territory; the specifically causal character of the resulting models is explained by the shared features of the observers (temporal embedding, intervention capacity, bounded processing).

3.3 The Self-Undermining Problem

The eliminativist thesis states: ‘bounded observers construct causal models *because of* their limitations.’ But ‘because of’ is itself a causal locution. The thesis uses causation to explain away causation. If one deflates this meta-level claim as also ‘just compression,’ the thesis ceases to be *true* and becomes merely *useful*—forfeiting its status as a discovery about reality.

The perspectival account avoids this regress. Because causation is real at the relational level (not fictitious), the meta-level causal claim—‘boundedness causes causal modeling’—is a legitimate, relationally real claim. The account is self-consistent: it uses relational causation to explain the emergence of relational causation, without requiring the kind of absolute, observer-independent causation it denies.

4. Relationship to Existing Perspectivalism

4.1 Price’s Agent Causation

Price (2007) and Price and Ramsey (2007) develop an explicitly agent-based perspectivalism: causal judgments are projections of the agent’s deliberative perspective onto the world. Our account is sympathetic but differs in two respects. First, we ground the observer’s contribution in information-theoretic compression rather than in agency and deliberation. This is broader: any bounded information-processing system—including a thermostat or a simple control loop—can be said to ‘construct’ a causal model in the relevant sense. Agency is sufficient but not necessary for perspectival causation. Second, we are more explicit about the territory’s contribution. Price sometimes gives the impression that causation is entirely agent-contributed; we insist that the undirected conditional independence structure is observer-independent and constrains which causal models are viable.

4.2 Ladyman and Ross’s Structural Realism

Ladyman and Ross (2007) argue that the world’s fundamental ontology consists of structures and patterns, not objects with intrinsic natures. Our account fits naturally within this framework: the conditional independence structure of the world is a ‘real pattern’ in Dennett’s (1991) sense, and causation is a higher-order pattern that emerges when bounded observers compress the first-order pattern. Structural realism provides the metaphysical backstop that prevents the perspectival account from collapsing into idealism: the structure is real and observer-independent, even though its causal interpretation is not.

4.3 Woodward's Interventionism

Woodward (2003) provides the most developed interventionist account of causation. Our relationship to Woodward is complex: we accept his formal apparatus while reinterpreting his ontological commitments. We agree that invariance under intervention is the correct criterion for distinguishing causal from merely correlational relationships. But we argue that the intervention framework itself presupposes an observer-contributed decomposition of the world into variables, system boundaries, and temporal direction (§2.4). Woodward's 'invariance' is therefore real relative to a perspective, not absolute.

A Woodwardian may respond that the intervention conditions pick out objective features of the world—that modularity, for instance, is a real physical property. We agree that modularity tracks something real (the factorization structure of the joint distribution), but note that the *identification* of modules requires a variable partition, which is observer-contributed. The objectivity of the territory's structure and the perspectivity of its causal interpretation are compatible—this is precisely the relational view.

4.4 Algorithmic Causality: Janzing and Schölkopf

The algorithmic independence of causal conditionals (Janzing and Schölkopf 2010) deserves special attention, as it may appear to ground causal direction in observer-independent algorithmic structure. We addressed this in §2.4, but add a further observation: the framework presupposes a bivariate setting (X, Y) with a pre-specified variable partition. The question 'does X cause Y or Y cause X ?' already assumes the observer has carved the world into X and Y . The algorithmic criterion then selects the orientation that yields the most compressible factorization—a compression-quality principle fully consistent with our account: the territory provides the dependence; the observer provides the variables; algorithmic independence arbitrates the direction within that observer-relative setup.

5. The Ontological Commitment: Disambiguated

The slogan 'no bounded observer, no causation' is ambiguous between at least three readings:

(a) Model-dependence. Without observers, there are no *causal models*. This is trivially true and philosophically uninteresting.

(b) Fact-dependence. Without observers, there are no *causal facts*. This is bold but far stronger than anything the argument establishes, and risks collapsing into idealism.

(c) Direction-dependence. Without observers, the world contains undirected conditional independence structure but no *directed causal facts*. The directional, scale-specific, boundary-relative content we call 'causation' requires an observer's perspective.

We defend (c). The territory is rich with real structure—conditional independencies grounded in physical locality, conservation laws, and symmetries. This structure is fully objective. But the *directed* causal graph— X causes Y rather than Y causes X —requires specifications that the territory alone does not supply: a coarse-graining, system boundaries, and a temporal arrow. These are the observer's contribution.

On reading (c), a world without observers would still contain dependency structure but would lack causal directionality—just as a world without macroscopic observers would still contain microphysical

states but would lack temperature. The analogy is precise: temperature requires a macrostate partition; directed causation requires an observer's perspective.

6. Consequences

6.1 Causal Pluralism Becomes Mandatory

The same physical system legitimately admits different causal descriptions at different scales. A neuroscientist and a quantum physicist are not discovering the *same* causes—they are constructing different compressions of the shared underlying conditional independence structure, using different variable partitions and system boundaries. This is not a failure of science; it is the expected outcome once we recognize that causal structure is partly observer-contributed.

Hoel, Albantakis, and Tononi (2013) provide formal support: their effective information measures demonstrate that macro-level causal descriptions can carry more causal information than micro-level descriptions of the same system. This is inexplicable on a view where causation is a unique, scale-independent feature of the territory; it is precisely what the perspectival account predicts.

6.2 The Fundamental Level Loses Metaphysical Privilege

If causation is relational, microphysics is not 'more causally real' than macroeconomics—it is a different compression at a different scale, with different variables and boundaries. The claim 'the real causes are at the level of particles' presupposes a scale-independent notion of 'real cause' that the perspectival account denies. This aligns with the emergentist arguments of Anderson (1972) and Batterman (2002): higher-level explanations are not mere approximations to lower-level ones but autonomous descriptions with their own domain of causal validity.

6.3 Mental Causation: Partial Dissolution with Caveats

The causal exclusion problem (Kim 1998, 2005) holds that if the physical is causally closed and mental properties are distinct from physical properties, mental causation is either epiphenomenal or reducible. The argument depends on the assumption that causal relations at different levels compete for the same causal 'slot.'

The perspectival account weakens this assumption. If causal structure is partly observer-contributed, then 'physical cause' and 'mental cause' are compressions of the same underlying dependency structure at different coarse-graining levels. They do not compete because they are not answers to the same causal question—they address different scales with different variable partitions.

However, we acknowledge this does not fully dissolve the exclusion problem. Kim's argument has a specific logical structure: if physical events have sufficient physical causes, and mental events are not identical to physical events, then mental events are causally redundant. Our account challenges the premise that 'having a sufficient physical cause' is a scale-independent claim—on our view, sufficiency is relative to a variable partition. But fully resolving the exclusion problem would require showing that there is no principled partition at which the physical compression screens off the mental one. This is a substantive empirical and formal question—specifically, whether the effective information at the mental coarse-graining

exceeds that at neurophysiological grainings, in the sense of Hoel et al. (2013). We flag this as an open question rather than claiming resolution.

7. The Residual Hard Question

Why is there conditional independence structure at all? Why does the joint distribution over the world’s variables factorize so neatly? If causation is compression of correlations, something about reality must explain why reality is *compressible*—why the universe exhibits sparse dependency structure rather than being one holistic entangled system.

One answer: the sparsity is a brute fact. A deeper answer connects it to *locality in physics*—interactions fall off with distance, which guarantees approximate conditional independence between spatially separated subsystems. Locality, together with the symmetries encoded in the Standard Model and general relativity, is arguably the real foundation that makes causal modeling possible—and it is a feature of the territory, not of the observer.

The final picture:

The territory provides sparse correlational structure via locality and symmetry. Bounded observers compress that structure into directed graphs. The compression is the causation. Neither the observer nor the territory alone produces it—it is genuinely relational.

8. Toward Formalization

The perspectival account suggests a research program: formalizing the ‘quality’ of a causal compression. One seeks a metric evaluating which observer partitions yield the most stable, portable, intervention-supporting causal graphs. Hoel, Albantakis, and Tononi’s (2013) effective information measures represent a promising beginning, as does the causal abstraction framework of Beckers and Halpern (2019) and the causal feature learning approach of Chalupka, Perona, and Eberhardt (2017).

A fully developed formalism would specify: (i) the space of possible observer partitions; (ii) a compression cost function over directed graphs; (iii) a stability criterion measuring how well a causal graph generalizes across interventions and environments; and (iv) a proof that the partitions maximizing (iii) relative to (ii) track the system’s conditional independence structure.

We note that these desiderata are non-trivial. Point (iv) in particular is demanding: it amounts to a general theorem connecting optimal compression with conditional independence, and may not hold in full generality. Preliminary results in specific settings—linear Gaussian models and discrete causal systems (Chalupka et al. 2017)—are encouraging, but a general proof remains open. Indeed, there are reasons to suspect that in highly entangled quantum systems, the factorization assumption may break down, which would mark a boundary condition on the applicability of causal modeling itself—a prediction the perspectival account makes naturally but standard causal realism does not. We regard the formal development of (i)–(iv) as the most important open question the perspectival account generates. The formal conditions under which closed dependency structures are coherent—including fixed-point existence and topological closure—are developed independently in Kriger (2024).

9. Conclusion

Causation, we have argued, is neither an illusion nor a mind-independent feature of ultimate reality. It is a relational structure—emergent, frame-dependent, and genuinely real at the observer–system interface. The territory provides the raw material: undirected conditional independence structure grounded in physical locality and symmetry. The bounded observer provides the directed compression: a causal graph that renders the environment legible for prediction and intervention. Neither component alone constitutes causation; together, they do.

This reframing does not destroy science—it clarifies its nature. The scientific project is revealed as the search for maximally powerful causal compressions: variable partitions and directed graphs that optimally track the territory’s dependency structure relative to an agent’s intervention capacities and predictive needs. The distinction between this view and standard causal realism is subtle but consequential: it mandates causal pluralism, challenges the metaphysical privilege of microphysics, and reframes persistent puzzles like the problem of mental causation—without claiming to dissolve them entirely.

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